

Continuous functions – 2

Monotonicity, oscillation and the horizontal chord

Homework

1 сентября 2026 года

Theoretical minimum. The horizontal chord theorem and the difference trick $g(x) = f(x + \alpha) - f(x)$; continuity together with injectivity implies strict monotonicity; oscillation at a point; the range of a continuous function on an interval; iterates of a continuous self-map of a segment; local extrema and monotonicity.

1. Define the oscillation of $f: \mathbb{R} \rightarrow \mathbb{R}$ at x_0 as the infimum, over all neighbourhoods U of x_0 , of $\sup_U f - \inf_U f$. Suppose $\text{osc } f(x) = 1$ for every $x \neq 0$. Find $\text{osc } f(0)$.
2. Does there exist a continuous $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $\lim_{x \rightarrow x_0} \frac{|f(x) - f(x_0)|}{|x - x_0|} = +\infty$ at every point x_0 ?
3. (IMC 2006) Let $f: \mathbb{R} \rightarrow \mathbb{R}$. Prove or disprove:
 - a) if f is continuous and $\text{range}(f) = \mathbb{R}$, then f is monotone;
 - b) if f is monotone and $\text{range}(f) = \mathbb{R}$, then f is continuous;
 - c) if f is monotone and continuous, then $\text{range}(f) = \mathbb{R}$.
4. (IMC 1996) Prove that for a continuous $f: [0, 1] \rightarrow [0, 1]$ the sequence of iterates $x_{n+1} = f(x_n)$ converges if and only if $x_{n+1} - x_n \rightarrow 0$.
5. (IMC 2023) Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ with a continuous second derivative for which $f(7x + 1) = 49f(x)$ for all $x \in \mathbb{R}$.
6. (IMC 1997) Say that a continuous $f: [0, 1] \rightarrow \mathbb{R}$ *crosses the axis* at x if $f(x) = 0$ while every neighbourhood of x contains y, z with $f(y) < 0 < f(z)$.

- a) Give an example of a continuous function crossing the axis infinitely often.
 - b) Can a continuous function cross the axis uncountably often?
7. (RMO 2002) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ have a limit at every point and no local extremum. Show that a) f is continuous; b) f is strictly monotone.
8. (RMO 2005) Let $f: [0, 1) \rightarrow (0, 1)$ be continuous and surjective.
- a) Prove that for every $a \in (0, 1)$ the restriction $f|_{(a, 1)}: (a, 1) \rightarrow (0, 1)$ is surjective.
 - b) Give an example of such a function.
9. (RMO 2006) Let $f: [0, \infty) \rightarrow \mathbb{R}$ be such that for every $x > 0$ the sequence $\{f(nx)\}_{n \geq 0}$ is increasing.
- a) If f is continuous on $[0, 1]$, must f be increasing?
 - b) The same question if f is continuous on $[0, \infty)$.
10. (RMO 2007) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous and let $a < b$ be two points of the image of f . Show that there is a closed interval I with $f(I) = [a, b]$.